A CONDITIONAL SKEIN-LASAGNA OBSTRUCTION FOR A TRACE-73 CAPPELL–SHANESON CANDIDATE

ANONYMOUS REPOSITORY AUDIT

Abstract. We study a proposed trace-73 Cappell–Shaneson construction associated with the matrix

$$A = \begin{pmatrix} 0 & 269 & 1240 \\ 0 & 41 & 189 \\ 1 & 0 & 32 \end{pmatrix}.$$

A public finite calculation produces the nonzero cubic Burau scalar -59072 in the grading convention corresponding to quantum degree 494. An exact Artin–Magnus certificate and the pure-braid Andreadakis theorem establish a third-order property of the public braid. We give an AR-side product-annulus model and compute the finite cubic scalar. Conditional on the embedded handle presentation, the actual MWW coefficient maps, and the three-handle comparison, the degree-494 class would obstruct a diffeomorphism to S^4. The Johnson-generator lift supplies a candidate replacement collar and exact six-sweep word, but the paper does not identify it with the historical point-push geometry. The Lean development checks the finite and quotient algebra but does not construct the external structures.

Contents

1.	Introduction	2
	Contributions	3
2.	The trace-73 Cappell–Shaneson candidate	3
3.	Precise statements	4
4.	The finite computational layer	5
4.1.	Integral lattice calculations	5
4.2.	The public Burau computation	6
5.	Lasagna modules and the intended obstruction	6
6.	Johnson product-ribbon presentation and P0	7
6.1.	The actual AR link	7
6.2.	Return to the linear torus map	8
6.3.	Words and normal fields	8

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6.4. The two cancellations 9
6.5. The actual detector ball 9
6.6. Failure of the earlier symbolic lifts 10
6.7. P0 conclusion 13
7. The coefficient-trace comparison 13
7.1. The product Hattori equivalence 14
7.2. Quantum trace and the endpoint map 15
7.3. Selected class and divided detector 15
7.4. The complete two-handle cocone 16
7.5. C conclusion and the degree-494 square 18
8. Three-handle sphere closure 18
8.1. The remaining essential-sphere endpoint comparison 20
9. Final identifications 20
10. Formalization boundary 20
11. Status table and reproducibility ledger 21
12. Related work 22
13. Reproducibility and review 23
14. Conclusion 23
Appendix A. Retired historical objects 24
Appendix B. Appendix B: strict P0 reconstruction protocol 24
Appendix C. Correspondence with the Lean fields 26
References 26

1. Introduction

The smooth four-dimensional Poincare conjecture asks whether every smooth closed manifold homotopy equivalent to S^4 is diffeomorphic to S^4. Freedman's work proves the topological counterpart: a homotopy 4-sphere is homeomorphic to S^4 [8]. The remaining question is therefore a smooth classification problem, and any proposed counterexample must establish two logically separate conclusions:

(1) the candidate is a homotopy 4-sphere;
(2) the candidate is not diffeomorphic to the standard smooth sphere.

Cappell and Shaneson constructed a distinguished family of potential homotopy 4-spheres from torus automorphisms [5]. Their handle structures were studied by Aitchison and Rubinstein [1]. Many subfamilies were subsequently shown to be standard, through work including Akbulut [2], Gompf [9], Kim–Yamada [13], and Iwaki [11]. This history makes geometric identification especially important: agreement of a matrix, a relator, or a trace parameter is not a substitute for identifying a complete framed handle presentation.

Skein lasagna modules extend Khovanov–Rozansky link homology to smooth four-manifolds [17]. Manolescu and Neithalath gave a two-handlebody description [15], and Manolescu, Walker, and Wedrich gave formulas compatible with general handle decompositions [16]. These results provide an attractive route to conclusion (ii): a nonzero class in a nonzero bidegree cannot be carried by a diffeomorphism to the lasagna module of S^4, which is concentrated in bidegree zero.

We investigate the public finite class for the standard-form Cappell–Shaneson parameter $(41, 189, 73)$. Compact product-ribbon, point-push, owner-lift and Nielsen ledgers replace parts of the unavailable historical data. A Johnson-generator replacement supplies a candidate detector word, while the actual handlebody identification, MWW coefficient maps, and sphere replacement fixed near the detector remain hypotheses.

Contributions. The main contributions are as follows.

(1) We construct a splitting-preserving Johnson lift and an exact six-sweep 44-channel word equal to the public 11340-letter word; its identification with the historical AR point-push remains a premise.
(2) We isolate the two-sided Hattori and all-state beta/psi maps required for C, but retain their candidate-level identification as hypotheses.
(3) We use HJ to state the relative sphere-system problem and isolate the additional fixed-detector and essential-sphere endpoint maps required for S.
(4) We provide executable ledgers, mutation tests and a kernel-checked Lean algebraic core.

Historical files referenced by earlier versions of the project are not publicly available. Their hashes are retained as provenance and are not replaced by narrative JSON fields or terminal status strings.

2. The trace-73 Cappell–Shaneson candidate

Let

$$A = \begin{pmatrix} 0 & 269 & 1240 \\ 0 & 41 & 189 \\ 1 & 0 & 32 \end{pmatrix}. \tag{1}$$

This is Iwaki's standard form $X_{41,189,73}$; the triple occurs among the trace-73 representatives in his table [11, Table 2]. Direct integer arithmetic gives

$$\det A = 1, \qquad \det(A - I) = 1. \tag{2}$$

For $A \in \mathrm{SL}(3, \mathbb{Z})$, let $f_A : T^3 \to T^3$ be the induced diffeomorphism, made equal to the identity near a chosen point. Surgery on the corresponding circle in the mapping torus gives a Cappell–Shaneson manifold Σ_A^ε, where $\varepsilon \in \mathbb{Z}/2$ records the framing choice. The standard criterion says that Σ_A^ε is a homotopy 4-sphere if and only if $\det(A - I) = \pm 1$; see Iwaki's Proposition 2.1 and the original sources cited there [11, 5].

Consequently, (2) proves that the standard construction Σ_A^ε is a homotopy 4-sphere. It does not prove that a separately generated large Kirby diagram is a diagram of that construction. That identification is the first and most load-bearing hypothesis below.

Iwaki proves standardness for many infinite families. The appearance of $(41, 189, 73)$ in a representative table is not a standardness or exoticness theorem for this individual sphere. Likewise, the trace bound in Kim–Yamada's work stops below 73 [13]. These observations explain the choice of candidate but do not decide its diffeomorphism type.

3. Precise statements

We first state the abstract theorem in the form actually proved by the Lean development. This prevents geometric intuition from silently strengthening the result.

Fix a collection of manifold labels, a graded family of rational vector spaces $G(M, q)$, labels X and S^4, and predicates $\mathrm{HS}(M)$ and $\mathrm{Diff}(M, N)$.

Theorem 3.1 (Abstract nonstandardness theorem)**.** *Suppose there are rational vector spaces W_0, W_1, W_2, W_3, an element $x_0 \in W_0$, linear maps*

$$W_0 \xrightarrow{q_{01}} W_1 \xrightarrow{q_{12}} W_2, \qquad \ell_i : W_i \longrightarrow \mathbb{Q} \quad (i = 0, 1, 2),$$

and linear isomorphisms

$$T : W_2 \xrightarrow{\cong} W_3, \qquad F : W_3 \xrightarrow{\cong} G(X, 494),$$

such that

$$\ell_0(x_0) = -59072, \tag{3}$$

$$\ell_1 q_{01} = \ell_0, \tag{4}$$

$$\ell_2 q_{12} = \ell_1. \tag{5}$$

Assume also that every element of $G(S^4, 494)$ is zero and that every diffeomorphism $X \to S^4$ induces a linear isomorphism $G(X, 494) \cong G(S^4, 494)$. Then X is not diffeomorphic to S^4.

Proof. Set $v = F(T(q_{12}(q_{01}(x_0))))$. If $v = 0$, injectivity of F and T implies $q_{12}(q_{01}(x_0)) = 0$. Applying ℓ_2 and using (4)–(5) gives $-59072 = 0$, a contradiction. Thus $v \neq 0$. If X were diffeomorphic to S^4, the induced isomorphism would send v to an element of $G(S^4, 494)$, hence to zero, contradicting injectivity. □

The Lean theorem corresponding to Theorem 3.1 accepts the input structures `ExternalGeometry` and `CSExternalGeometry`; no inhabitants of these structures are constructed in Lean. We state the candidate-specific inputs here and prove them geometrically in Sections 6–9.

Hypothesis 3.2 (Embedded candidate presentation, P0)**.** The explicit Johnson-generator replacement is a labelled and framed handle presentation of Σ^0_A, with the two framed product cancellations, the embedded detector ball and the stated cable attachments.

Hypothesis 3.3 (Coefficient comparison, C)**.** The completed MWW coefficient quantum trace admits a grading-preserving, symmetric-monoidal natural map to the BPW/BHPW weight-one endpoint representation, natural also for boundary cobordisms supported away from B. Under this map the point-push action is the public Burau representation, and the divided cubic row descends through every beta and psi relation. Its selected value is -59072 in the public normalization; all strict pivotal sign variants are nonzero. The class lies in quantum degree 494.

Hypothesis 3.4 (Relative three-handle closure, S)**.** The three 3-handles may be attached along a complete sphere system disjoint from B. For each sphere the detector kills the undotted MWW relation and identifies the once-dotted map with the source term. Hence the divided detector descends through the complete three-handle coequalizer.

Hypothesis 3.5 (Final candidate identifications, P3)**.** MWW's three-/four-handle formulas identify the final quotient with $\mathcal{S}^2_{0,(0,494),0}(\Sigma^0_A;\mathbb{Q})$. The corresponding summand for S^4 is zero, and diffeomorphisms induce grading-preserving equivalences. Moreover, $\det(A-I)=1$ makes Σ^0_A a homotopy sphere.

Theorem 3.6 (Conditional trace-73 theorem)**.** *Assume Hypotheses 3.2–3.5. Then the trace-73 Cappell–Shaneson candidate satisfies*

$$\Sigma^0_A \simeq S^4 \quad \textit{and} \quad \Sigma^0_A \not\cong_{\mathrm{diff}} S^4.$$

Proof of Theorem 3.6. Hypothesis 3.2, Hypothesis 3.5, and the Cappell–Shaneson criterion give the homotopy-sphere assertion. Hypothesis 3.3 supplies the coefficient comparison, and Hypothesis 3.4 supplies the three-handle descent. They instantiate Theorem 3.1, whose selected class is nonzero in quantum degree 494. MWW's four-handle isomorphism preserves this degree, while the same degree of the standard S^4 module vanishes. Diffeomorphism invariance and Theorem 3.1 rule out a diffeomorphism to S^4. ☐

4. The finite computational layer

The public finite layer has three independent parts: lattice arithmetic, a chosen-sphere coordinate determinant, and a truncated Burau calculation.

4.1. **Integral lattice calculations.** Besides (2), the repository records the proposed sphere coordinate columns

$$C=\begin{pmatrix}-1311 & -189 & 41\\ 8608 & 1241 & -269\\ -1 & 0 & 1\end{pmatrix}, \qquad \det C = 1. \tag{6}$$

Thus the three recorded coordinate vectors form a basis of $\mathbb{Z}^3$. This is only the last arithmetic step in a geometric-basis argument. It does not construct embedded spheres, prove disjointness, or identify $\mathbb{Z}^3$ with $H_2^{\mathrm{sph}}(\partial W_2)$.

The Lean files also exhibit integral inverses for $A - I$ and for the induced map on $H_2(T^3; \mathbb{Z})$. These computations eliminate the candidate-specific lattice algebra from the homotopy-sphere branch. Van Kampen, Wang–Mayer–Vietoris, Poincare duality, and Hurewicz–Whitehead arguments remain supplied through a topology structure rather than formalized as four-manifold topology.

4.2. **The public Burau computation.** Let $\varepsilon = t - 1$. The script `scripts/recompute_t73_delta3.py` reads a result-free JSON input and performs the following finite operations:

(1) parse 252 primitive crossing rows and reconstruct an 11,340-letter Artin word on 44 strands;
(2) cable it to a 45,360-letter word on 88 strands;
(3) apply the unreduced Burau representation over $\mathbb{Z}[\varepsilon]/(\varepsilon^7)$;
(4) substitute $t = q^{-2}$ with $q = 1 + h$, equivalently $\varepsilon = (1 + h)^{-2} - 1$;
(5) extract the coefficient of h^3 after the specified source vector and covector are applied.

Denote the resulting scalar by δ_3^{Bur}. The independently reproduced public result is

$$\delta_3^{\mathrm{Bur}} = -59072 = (-2)^3 \cdot 7384. \tag{7}$$

The public input does not contain either decimal literal 7384 or -59072; the former is, however, stored as the constant `cubicBase` in the Lean finite layer. Lean checks only the arithmetic $(-2)^3 \cdot 7384 = -59072$ and never evaluates the 45,360-letter word.

Proposition 4.1 (Exact scope of the computation)**.** *Equation* (7) *is a statement about the specified truncated Burau model. The script and JSON alone do not define the MWW invariant. Under the coefficient-trace comparison of Theorem 3.3, however,*

$$\delta_3^{\mathrm{MWW}} = \delta_3^{\mathrm{Bur}} = -59072,$$

up to the harmless global convention unit normalized in that theorem.

Proof. The script defines vector and covector operations in an 88-dimensional truncated polynomial representation. The displayed MWW equality is exactly the additional content of Hypothesis 3.3; it does not follow from the finite script or from the local Burau-block calculation. □

Thus the finite recomputation and the categorical comparison have distinct roles: the former supplies the integer, while the latter makes it a four-manifold invariant.

5. Lasagna modules and the intended obstruction

We recall only the features needed for the comparison. For a smooth compact oriented four-manifold W and a framed link $L \subset \partial W$, the skein lasagna module $\mathcal{S}_0^N(W;L)$ is generated by decorated surfaces in W, modulo local relations coming from KhR_N cobordism maps [17, 16].

For a handle decomposition

$$W_1 \subset W_2 \subset W_3 \subset W_4,$$

where W_1 is a one-handlebody, W_2 adds two-handles along a framed link, W_3 adds three-handles along embedded spheres, and W_4 adds four-handles, the published results have the following roles.

- MWW Theorem 3.2 identifies the two-handle stage with a cabled skein lasagna quotient, whose relations include beta and psi operations.
- MWW Theorems 3.7 and 3.10 describe each three-handle as a coequalizer of the two hemisphere maps, and give the complete handle formula.
- MWW Proposition 3.4 says that a four-handle attachment induces an isomorphism for the empty boundary link.
- MWW Corollary 3.5 gives $\mathcal{S}_0^N(S^4) \cong \mathbb{Z}$, concentrated in bidegree $(0,0)$.

At $N=2$, a nonzero class in bidegree $(0,494)$ would therefore obstruct a diffeomorphism to S^4. The intended proof constructs a functional at the one-handle stage, shows that it annihilates the two-handle and three-handle relations, and transports the resulting nonzero class through the four-handle isomorphism.

The formal algebra correctly captures the last sentence. The next sections prove the finite parts of P0 and C and isolate the additional geometric identifications required for P0, C and S.

6. Johnson product-ribbon presentation and P0

We first record the AR-side construction and the failure of earlier compact lifts, then give an explicit Johnson-generator replacement proving Proposition 3.2. The construction does not recover or use the unavailable historical planar diagram.

6.1. The actual AR link. Use the coordinate-spine genus-three Heegaard splitting of T^3 from Aitchison–Rubinstein [1, pp. 5–12]; the complete volume is available in a public Berkeley scan. Let C_i be the three coordinate spine circles with common base vertex Q, and let A_i be pairwise-disjoint narrow disk neighborhoods away from Q. Write t for the mapping-torus one-handle. If ψ_A is the handlebody-preserving representative supplied by their Lemma 3.1, the core of the ith mapping-torus two-handle is the embedded circle

$$C_i^- \cup \lambda_i \cup \psi_A(C_i)^+ \cup \mu_i, \tag{8}$$

where λ_i, μ_i are the two arcs through t. The framing annulus is constructed simultaneously as

$$A_i^- \cup (\lambda_i \times I) \cup \psi_A(A_i)^+ \cup (\mu_i \times I). \tag{9}$$

The disks, cones and product levels in the AR construction are pairwise disjoint, so (8)–(9) define the whole embedded framed link, not merely its words.

The remaining three fiber two-handles are the standard dual two-cells of the coordinate-spine splitting. After choosing orientations, denote their boundary curves by

$$r_{xy} = [x, y], \qquad r_{yz} = [y, z], \qquad r_{zx} = [z, x].$$

Their embeddings are defined by the dual cells and their framings are the fiber product framings. The untwisted section-surgery handle h_{CS} goes geometrically once over t. Thus the starting handle counts are $(1, 4, 7, 3, 1)$ in indices $0, \dots, 4$: AR p. 8 removes one 3- and the 4-handle from the mapping torus and glues $S^2 \times D^2 = H^2 \cup H^4$ in their place.

For an independently checkable standard-form bridge, set

$$C_{AR} = \begin{pmatrix} 0 & 0 & 1 \\ 713 & 83 & 0 \\ -902 & -105 & -10 \end{pmatrix}, \qquad R = \begin{pmatrix} 71 & -466 & 1 \\ -610 & 4004 & -7 \\ 1 & -7 & -2 \end{pmatrix}.$$

Direct multiplication gives

$$\det R = 1, \qquad C_{AR}R = RA,$$

and both matrices have determinant one and determinant-one difference from the identity. Hence the fiber diffeomorphism induced by R transports the complete AR construction to the displayed matrix A. Since $R \in GL^+(3, \mathbb{R})$ and this group is connected, its derivative preserves the homotopy class of the canonical normal framing of the section circle; in particular, the untwisted choice $\varepsilon = 0$ is preserved.

6.2. **Return to the linear torus map.** Let ϕ_A be the linear torus map, straightened to the identity on the section ball, and choose an isotopy ρ_u, $u \in [-1, 1]$, from ϕ_A to ψ_A, stationary near its endpoints and relative to that ball. Put $g_u = \rho_u \phi_A^{-1}$. With mapping-torus convention $(x, -1) \sim (\phi(x), 1)$, the formula

$$F([x, u]_{\phi_A}) = [g_u(x), u]_{\psi_A} \tag{10}$$

defines a diffeomorphism. Indeed, $g_{-1} = \mathrm{id}$, $g_1 = \psi_A \phi_A^{-1}$, and therefore $\psi_A g_{-1} = g_1 \phi_A$, which is precisely well-definedness at the seam. Endpoint stationarity makes (10) smooth, and the family g_u^{-1} gives its inverse. It fixes the section ball. Pulling the entire AR handle decomposition back by F replaces the top cores by the linear images $\phi_A(C_i)$ and transports every component and framing annulus simultaneously.

6.3. **Words and normal fields.** Lift C_i and $\phi_A(C_i)$ to the universal cover. The three top lifts begin at the same point $A(Q)$. Translate the whole top fiber by $-A(Q)$ and extend this translation over its collar. It is isotopic to the identity and transports the entire link and its normal fields. Since each attaching core is unbased, changing the first face crossing only cyclically permutes its word. We may therefore take the top centerline to be the straight segment from 0 to Ae_i; it crosses the jth integer faces at the rational times $k/(Ae_i)_j$. A fixed arbitrarily small product perturbation orders simultaneous events. Reading the two base arcs and the oppositely oriented bottom core gives exactly

$$m_i = t\phi_A(x_i)t^{-1}x_i^{-1}. \tag{11}$$

This is the full event rule used by the public generator.

The framing is equally explicit. AR push the coordinate axes in the universal cover in direction $(1,1,1)$; the resulting strips have parallel boundary components, and linearity of A preserves this property on their images [1, pp. 16–17]. Together with the λ_i/μ_i rectangles, these strips are the normal fields in (9). Thus no blackboard integer is substituted for a framing.

6.4. **The two cancellations.** Aitchison–Rubinstein identify (t, h_{CS}) as a complementary geometric one/two-handle pair [1, p. 7]. Slide every other two-handle passage over h_{CS} along its product rectangle and cancel; this removes the t, t^{-1} passages with zero relative twist.

Now $Ae_1 = e_3$, so the actual curve m_1 becomes zx^{-1} and has one geometric passage over the x one-handle. Thus (x, m_1) is another complementary pair. Sliding every other x passage over m_1 along parallel product bands replaces x by z and cancels the pair. All labels and normal fields move with the bands. The resulting counts are $(1,2,5,3,1)$, and the exact word projection is

$$|m_2| = 311, \quad [m_2]_{(y,z)} = (40, 269),$$
$$|m_3| = 1460, \quad [m_3]_{(y,z)} = (189, 1271).$$

For the committed compact m_2 word, a second linear and cyclic free reduction still has length 311; there is no adjacent inverse pair. The explicit 19-step Nielsen representative constructed in Appendix B instead reduces to length 309. It contains 40 y/Y passages, rather than 42; with the two r_{xy} passages its detector has 42 channels rather than 44. Thus the 309/311 discrepancy is a genuine representative discrepancy, not merely a printing convention. The transported r_{zx} has empty free word, but no split-unknot conclusion is needed or inferred from that fact.

6.5. **The actual detector ball.** Cut the remaining genus-two handlebody along its y, z disk system. The selected m_2/r_{xy} state has 42 y passages from m_2 and two from r_{xy}. At distinct product-normal levels, a regular neighborhood of these 44 passages is a standard three-ball B containing a punctured-disk collar. The six affine sweeps specify a pure 44-strand braid in this collar. The mapping-class interpretation of the braid group realizes it by an isotopy of the punctured disk, and isotopy extension gives an ambient isotopy of B. Choose the extension to be the identity to first order near every returned puncture at its terminal time. Hence each product normal returns, not merely the total writhe. The motion acts on every oriented normal copy simultaneously. The public verifier regenerates all 252 factors and obtains

$$|B_{44}| = 11340,$$

$$\mathrm{SHA256}(B_{44}) = \begin{array}{l}\texttt{7C2D2F792C2672221A76CAF08A71F560A} \\ \texttt{F0CB7654B4D537C00FAD00B16EFA187}\end{array}.$$

$$\mathrm{perm}(B_{44}) = 1, \qquad \mathrm{wr}(B_{44}) = 0.$$

The symbolic construction, source digest, matrix bridge, handle counts, cancellation data and collar labels are regenerated in `audit/t73_ar_product_witness.json`. Its digest is

`AA9328EA3C7A53B937C97A02D611850D`
`0E8EEAF87C68756C04B013A486F6F112`.

The source-PDF, regeneration and mutation checks are described in the repository README.

6.6. **Failure of the earlier symbolic lifts.** The preceding symbolic formulas defined an AR-side handle model and an abstract pure braid in a standard punctured disk. By themselves they did not provide a labelled ambient isotopy or Kirby-move sequence placing the 44 selected passages in one specified framed 3-ball in the boundary of the actual reduced handlebody. In particular, the assertion at the start of the detector-ball construction that these passages have the required common regular neighborhood is an additional geometric input. The JSON witness records this assertion and its hash, but a SHA-256 digest proves only byte identity, not embeddedness or framing compatibility. This was the P0 gap in the earlier witness.

There is a sharper provenance obstruction. The 252 public crossing rows come from the unavailable frozen planar diagram; the compact point-push script regenerates those rows but does not derive them from the AR component parametrizations or cancellation bands. Mapping-class realization and isotopy extension construct a chosen pure braid in a standard ball. They do not prove that the chosen braid is the relative-endpoint isotopy class of the 44 pre-existing AR passages: that relative class is exactly the braid-group datum to be proved. Thus the earlier witness did not identify the public 11340-letter word as an AR collar braid.

The exact comparison in `scripts/compare_t73_nielsen_passages.py` therefore falsifies the current 19-step Nielsen representative as the source of the public 44-channel collar: it produces 42 channels. This does not falsify Hypothesis 3.2 globally, because a different handlebody-preserving AR representative may be related to the compact 311-letter representative. The Johnson construction below supplies such a replacement directly.

There is also a necessary free-basis test which the compact straight-word lift fails. If ψ_A preserves the genus-three handlebody H_B, then $\psi_A|_{H_B}$ induces an automorphism of $\pi_1(H_B) \cong F_3$; in particular, the three images of the spine generators form a free basis. GAP 4.12.1 [19] reports that the compact words of lengths $1, 310, 1461$ define an injective but non-surjective endomorphism of F_3. The same test reports the explicit Nielsen map to be bijective. Hence the compact straight words cannot themselves be the images of a handlebody-preserving ψ_A.

An IA correction supplies a new candidate: conjugating all three Nielsen images by x^{-1} preserves the abelianization A, remains an automorphism of F_3, and after the registered cancellations gives an m_2 word of length 311 with 42 y/Y passages, hence 44 channels together with r_{xy}. It is not the exact compact word. Its exact word movie to the compact representative contains 11,754 r_{yz} commutations and two free bigons, with net oriented r_{yz} coefficient -40. The embedded IA conjugation, band movie and framing comparison would still be required for that discarded route.

This IA correction does not itself construct a new unbased geometric representative. Simultaneously conjugating all three generator images changes the based $\mathrm{Aut}(F_3)$ lift but leaves the $\mathrm{Out}(F_3)$ class unchanged; each cyclically reduced spine image remains in the same conjugacy class. The two additional word passages are therefore basepoint-whisker data unless an embedded movie makes them actual collar passages. The finite audit verifies that the based count changes from 42 to 44 while all three unbased cyclic classes remain unchanged. Thus the x^{-1} correction is not an embedded P0 witness.

The search was therefore extended to non-inner IA automorphisms. A strict sufficient extension test requires the correction to carry the three dual meridians $[x, y], [y, z], [z, x]$ to a signed permutation of their conjugacy classes. This condition supplies an explicit disk-system extension over the complementary handlebody, though it is not necessary for every possible extension. The complete depth-three search visits 11,869 IA states and finds 751 genuinely detector-changing 44-channel candidates, but none pass this sufficient dual-meridian test. A depth-four run capped at 50,000 states is likewise negative but truncated. These bounded results eliminate the tested simple extensions; they do not prove global nonexistence.

A more appropriate search follows Johnson's explicit generators for the mapping class group of the standard genus-three splitting of T^3 [12]. His generator α_{ij} pushes the diagonal of the x_i–x_j square to a neighborhood of the spine, following one edge and then the other; pushing to the opposite side gives a second splitting-preserving lift of the same transvection. We factor A into 93 unit transvections and search these two side choices. The recorded 93-bit choice is independently reproduced by a deterministic seeded search and has the following exact properties:

$$|m_2| = 311, \qquad \#(y/Y) = 42, \qquad p_y = 42 + 2 = 44.$$

GAP verifies that the three resulting spine images form a free basis, and the reduced m_2 word agrees letter-for-letter with the compact word. Its class-two r_{yz} coefficient relative to the compact word is zero, so this route has no residual linking-dependent framing correction. The other two spine images differ from the non-basis compact straight words; this is how the new lift avoids the GAP obstruction.

For each of the 93 square moves the repository records the diagonal, the chosen two-edge path and a nonzero square normal. Truncating both paths in basis coordinates at one-quarter and three-quarters makes the movie relative to the spine vertex. If M ranges over the 93 intermediate unimodular bases, the uniform physical radius

$$r = \frac{1}{8\max_M \|M^{-1}\|_\infty} = \frac{1}{196104}$$

is fixed throughout. This supplies a splitting-preserving relative spine movie.

The reduced y one-handle has a standard product chart $D^2 \times [-1, 1]$. Inside it take

$$B = D^2 \times [-1/2, 1/2].$$

The exact Johnson m_2 word has 41 positive y passages followed by one negative passage; the standard $r_{xy} = [z, y]$ contributes one positive and one negative passage. Assigning the 44 passages distinct rational points in D^2 gives pairwise-disjoint product arcs in B. Wickets 1 and 2 belong to r_{xy}, wickets 3 through 44 are the ordered m_2 passages, and wicket 44 is its unique negative passage. The constant transverse product direction, with push size 1/10000, gives their framing.

The oriented dual rectangle for r_{xy} has six point-push legs. Reading the actual lane order and the orientation of wicket 44 gives 42 factors on each leg. Expanding these 252 geometrically defined L/R pure factors gives an Artin word of length 11340. Only after this AR-side construction is complete does the verifier read the public target; the two words agree letter-for-letter. A rational PL realization on the same lane points is then re-extracted from its projected crossings, returns all 44 endpoints, and gives the same word. Reversing the orientation of wicket 44 changes the word, providing a mutation check. Product translations near each moving puncture may be chosen with identity endpoint derivative, so every product normal returns.

Retired 309-word comparison. At the word level this comparison is now finite and exact. The two words collect to the same normal form $y^{40}z^{269}$; concatenating the two collection movies gives 11,257 commutations using the r_{yz} relator and one free bigon. The repository binds all 11,258 steps to four oriented local PL band templates and preserves the owner label m_2. What remains is to embed those bands in the actual reduced $m_2 \cup r_{yz}$ link and verify the global product-framing transport.

The framing obstruction reduces to one integer. The net oriented r_{yz}-slide coefficient of the complete movie is one, and the standard fiber handle has product framing $f(r_{yz}) = 0$. Hence the handle-slide formula in Kirby calculus [14] gives

$$f_{\text{final}}(m_2) - f_{\text{initial}}(m_2) = 2\,\text{lk}(m_2, r_{yz}).$$

Thus this movie preserves the required framing if and only if the actual reduced Kirby diagram has $\text{lk}(m_2, r_{yz}) = 0$. That linking integer is not determined by the word ledger and would remain an additional input for this retired comparison.

The extractor `scripts/extract_t73_ryz_linking.py` computes this integer from a labelled reduced-link PD ledger by halving the signed sum of mixed m_2–r_{yz} crossings and requires the normal-field transport receipt. The actual reduced PD ledger is absent at the current revision.

This absence cannot be repaired from the committed word/framing ledger. Indeed, two explicit local-clasp controls with identical component words and the same recorded component framings have mixed crossing sums 0 and 2, hence linking numbers 0 and 1. The script `scripts/falsify_t73_linking_from_words.py` verifies both ledgers with the same extractor. Thus the assertion that the current word data determine $\mathrm{lk}(m_2, r_{yz})$ is false; an embedded reduced-link diagram or equivalent normal-field movie is logically necessary.

This linking obstruction applies only to the retired 309-word comparison. The selected Johnson side-choice lift agrees with the compact m_2 word exactly and has relative r_{yz} coefficient zero, so no such slide or linking input occurs in the P0 proof.

6.7. **P0 conclusion.**

Proposition 6.1 (Scope of the Johnson certificate)**.** *The Johnson certificate proves the matrix factorization, free-basis test, relative square movie, rational lane construction, word-level cancellations, and equality of the independently generated six-sweep word with the public 11340-letter word for the explicit replacement presentation. It does not by itself prove P0a–P0c: the relative identification with the AR handlebody, complete framed Kirby cancellation, or compatibility with the MWW framing.*

Proof. Johnson's α_{ij} moves preserve the standard genus-three splitting. The exact transvection factorization and the recorded side choices give a representative of the matrix A. The relative square movie fixes the ball of radius $1/196104$ in the Johnson model. GAP verifies the resulting spine images form a free basis, while direct word replay gives $\psi_A(x) = z$ and the exact compact m_2 word. This verifies the algebraic part of the proposed cancellations; their ambient framed Kirby realization remains P0b.

The product chart of the surviving y handle supplies a candidate B and the 44 distinct rational lanes described above. Their owner and orientation data come from the replacement construction; identifying their framing with the AR/MWW framing is the separate P0c premise. The six-leg construction gives 252 pure factors, whose expanded and independently re-extracted Artin word agrees letter-for-letter with the public word; all endpoints and product normals return. These are precisely the certificate level assertions above. The historical PD is not used, and no historical-PD identity is claimed. □

7. The coefficient-trace comparison

We record the comparison required in C. The algebraic maps below are well-defined once the candidate-specific geometric identifications P0a–c and C1 are supplied; the numerical certificate is used only for the finite consequences of those maps.

7.1. **The product Hattori equivalence.** The compact words give

$$p_y = 42 + 2 = 44, \qquad p_z = 269 + 2 = 271.$$

More is true than this count. In the cyclic m_2 word, every one of the 42 y/Y events is followed immediately by a distinct z/Z event; the same holds for the two y/Y events of r_{xy}. The witness lists all 44 pairs. Under P0a–c, the corresponding one-edge intervals on the actual P0 product annuli are pairwise-disjoint subrectangles. Their removal leaves exactly 227 unpaired z passages, closing to separate product circles between the two parallel annulus boundaries.

Let $\mathcal{C}$ be the pregraded rational tangle category with objects $T : P_{86} \to P_{88}$, and let F be the framed west-to-east tangle formed by the 44 rectangles. MWW's one-handle theorem identifies the cut coefficient as

$$M_R(T, T') = \mathrm{KhR}_2(R \cup T' \cup \overline{T})$$

with its displayed shift and gluing actions [16, Theorem 4.7]. The fixed product isotopy, pivotal tangle duality, and Künneth over $\mathbb{Q}$ give graded isomorphisms

(12)
$$H_{T,T'} : M_R(T, T') \xrightarrow{\cong} \mathrm{Hom}_{\mathcal{C}}(FT, FT')\{-44\} \otimes A^{\otimes 227}, \qquad A = \mathbb{Q}[X]/(X^2).$$

If the product-ribbon identification is an actual framed isotopy, these maps form a coefficient-bimodule isomorphism. Indeed, gluing $f : S \to T$ before the source link and then applying the fixed product isotopy is isotopic rel boundary to first applying the isotopy and precomposing by Ff. The right action is the same argument with postcomposition. Thus both action squares are associativity of gluing for one simultaneous surface. At the chain level the two squares are

(17)
$$\begin{array}{ccc} C_R(T,T') & \xrightarrow{H_{T,T'}} & C(FT,FT') \otimes C(S^1)^{\otimes 227} \\ \downarrow L_f & & \downarrow L_{Ff} \otimes 1 \\ C_R(S,T') & \xrightarrow{H_{S,T'}} & C(FS,FT') \otimes C(S^1)^{\otimes 227}, \end{array} \qquad \begin{array}{ccc} C_R(T,T') & \xrightarrow{H_{T,T'}} & C(FT,FT') \otimes C(S^1)^{\otimes 227} \\ \downarrow R_g & & \downarrow R_{Fg} \otimes 1 \\ C_R(T,U) & \xrightarrow{H_{T,U}} & C(FT,FU) \otimes C(S^1)^{\otimes 227}. \end{array}$$

Here C_R and C are the strict Blanchet–Khovanov foam complexes. Both composites in each square are the same glued foam after exchanging two disjoint collars. BHPW strict functoriality removes the projective sign, and taking homology gives (12) and both MWW action squares. Thus (12) would be induced by the P0 product-ribbon isotopy, not inferred from the numbers 44 and 227. Constructing that actual isotopy and identifying the MWW KhR_2 complex with this foam complex is the C1/P0c premise.

Apply the 227 disjoint counits to (12). This is a homogeneous coefficient morphism. The remaining coefficient $(T, T') \mapsto \mathrm{Hom}(FT, FT')$ is representable by the invertible tangle F and hence is equivalent to the regular coefficient. The ordinary quotient statement, including both cyclic-relation spans, is kernel checked in `Smooth4PC/RepresentableCoefficient.lean`.

7.2. **Quantum trace and the endpoint map.** Set $R_q = \mathbb{Q}[q, q^{-1}]$. The pregraded cyclic relation is

$$L_f(m) = q^{|f|} R_f(m).$$

Conditional on C1, (12) and the counits are homogeneous coefficient morphisms, so they carry each such relation to the regular quantum vertical trace. BPW's canonical vertical-to-horizontal functor and the BHPW strict tangle/foam functor give

$$\mathrm{Sh}_q : q\,\mathrm{Tr}(\mathcal{C}; M_R) \longrightarrow \mathrm{Hom}_{R_q}(E_{86}, E_{88}), \tag{13}$$

after the flat-base qHH_0/Chern identification [4, 3]. Here

$$E_{86} = (V^{\otimes 86})_{\lambda=86}, \qquad E_{88} = (V^{\otimes 88})_{\lambda=86};$$

the first has rank one and the second rank 88. The tangle maps are $U_q(\mathfrak{sl}_2)$ intertwiners before restriction to these weight spaces.

Base change $q \mapsto 1 + h$ factors through rational localization and the completion of a Noetherian local ring, hence is flat. Reduction modulo h of the trace cokernel simply replaces $q^{|f|}$ by one, giving the ordinary MWW coefficient HH_0; right exactness of tensor product suffices for this last assertion.

7.3. **Selected class and divided detector.** Let $U : P_{86} \to P_{88}$ be the selected one-cup tangle in the P0 endpoint order and put $T_1 = F^{-1}U$. Define

$$v_T = H^{-1}_{T_1,T_1}(\mathrm{Id}_U \otimes X^{\otimes 227}).$$

The circle counits give $\mathrm{Sh}_h([v_T]) = u_h$. Its absolute degree is

$$-44 + 227 + 315 - 4 = 494.$$

The normalized checked R-matrix on adjacent one-defect basis vectors is

$$\begin{pmatrix} 1 - q^{-2} & q^{-1} \\ q^{-1} & 0 \end{pmatrix}.$$

After the position basis change and $t = q^{-2}$, it is the public positive Burau block; the negative block is its inverse. The physical cable has mixed orientations. BPW's oriented-cabling formula identifies it with the all-positive model up to a writhe-dependent grading shift and pivotal basis maps. The complete public word has writhe zero, so the global shift vanishes. Any degree-zero permutation or pivotal chart P transports the operator, cup and cap simultaneously:

$$W \mapsto PWP^{-1}, \qquad u \mapsto Pu, \qquad \ell \mapsto \ell P^{-1}.$$

The matrix coefficient is unchanged; we choose the public endpoint labels after this common transport. BPW's cup/cap formulas then imply that the constant terms, after the same position change, are

$$u_0 = e_0 \pm e_5, \qquad \ell_0 = e_{87}^* \pm e_2^*.$$

All omitted factors are invertible q-monomials or positive-order h-corrections.

Over $\mathbb{Q}[[h]]$ define

$$D_h = \ell_h(\rho_h(W) - I)\,\mathrm{Sh}_h\,.$$

This is well defined on the quantum coefficient trace because (13) has already imposed quantum cyclicity. The exact Magnus calculation and Darné's Andreadakis theorem give $W \in \Gamma_3(P_{44})$ [6, Theorem 6.2]; equivalently, and also by direct evaluation of all 7,744 entries,

$$\rho_h(W) - I \in h^3\,\mathrm{End}(E_{88}).$$

Thus D_h takes values in $h^3\mathbb{Q}[[h]]$. Division by h^3 is unique, and reduction modulo h gives a lift-independent ordinary functional

$$D_3 : HH_0(\mathcal{C}; M_R) \longrightarrow \mathbb{Q}.$$

Positive-order corrections in the cup, cap, and basis change do not alter the cubic coefficient.

The four possible strict pivotal signs give respectively

$$-53824, \quad -53760, \quad -59008, \quad -59072.$$

All are nonzero; the last is the public normalization. Hence

$$D_3([v_T]) = -59072 \neq 0 \tag{14}$$

in that normalization, without an unresolved sign premise.

7.4. **The complete two-handle cocone.** Only D_3 is asserted to descend. The P0 normal fields provide arbitrarily many disjoint parallel levels. We first type the construction at every MWW cable state. In owner order $(m_2, m_3, r_{xy}, r_{yz}, r_{zx})$, put

$$n_y = (42, 189, 2, 2, 0), \qquad n_z = (269, 1271, 2, 2, 0).$$

For $r \in \mathbb{N}^5$, cutting the actual cable $K(r,r)$ along the y- and z-cocores gives

$$p_y(r) = \sum_i r_i n_{y,i}, \qquad p_z(r) = \sum_i r_i n_{z,i}. \tag{24}$$

Under C1, parallel copies of the P0 rectangles give a chain isomorphism H_r of the same form as (17), with the corresponding endpoint sets. The split r_{zx} component is retained as a separate Frobenius factor. Since all copies lie in one fixed tubular product chart, H_r commutes with every gluing action and with adding a parallel pair.

Let $s_0 = e_{m_2} + e_{r_{xy}}$. If $r \geq s_0$, let Ω_r be the finite set of choices of one negative and one positive physical copy of each of m_2 and r_{xy}. For $\omega \in \Omega_r$, close every unselected product factor by the MWW core counit, use the selected 44 passages for the point-push, and write $D_{r,\omega}$ for the cubic coefficient of the resulting row. Define

$$D_r = |\Omega_r|^{-1} \sum_{\omega\in\Omega_r} D_{r,\omega}. \tag{25}$$

For a state not above s_0, add the missing pairs with the once-dotted MWW maps and define D_r by pulling (25) back. Disjoint supports make the order of these additions immaterial. Thus, conditional on C1, D_r is defined on every summand in MWW Definition 3.1, including the states below the selected one.

For beta, apply the P0 collar motion simultaneously to all physical copies. This is the oriented-cabling action of [4, Theorem E], made strict by BHPW. Constant permutations reindex Ω_r. After a placement is standardized, the remaining pure braid must be identified in every actual cable representation with $I + O(h)$ (C2). Simultaneous transport of the operator, cup and cap, together with $\rho(W) - I = O(h^3)$, changes the row only in order at least four. Therefore

$$D_r\beta_i(b) = D_r \tag{26}$$

for every $b \in B_{r_i,r_i}$. This cubic-order argument is exactly the C2 premise; it does not follow from the assertion that anti-parallel braid actions factor through a symmetric group, which MWW leave open [16, Remark 3.3].

For a pair addition, split the target placements according to whether the new pair is selected. A beta move exchanges a selected new pair with an old one, so (26) reduces both subsets to the same local core calculation. On the whole source that calculation is

$$(\epsilon\otimes\epsilon)\Delta(1) = 0, \qquad (\epsilon\otimes\epsilon)\Delta(X) = 1.$$

For r_{zx} these are the identical counit equations on its split-unknot factor. Forgetting the distinguished new pair has constant-size fibers between placement sets, and the normalizations in (25) cancel their cardinalities. The dotted raw degree $+2$ is cancelled by the target cabled shift -2. Consequently, for every owner and state,

$$D_{r+e_i}\psi_i^{[0]} = 0, \qquad D_{r+e_i}\psi_i^{[1]} = D_r. \tag{27}$$

Below s_0 the second equality is the definition; the Frobenius identities and disjoint-support interchange prove the first equality and the commutation of all owner squares.

Equations (26)–(27), together with the coefficient trace relation already handled by (17), prove on the whole cabled direct sum

$$D_3(\beta_i(b)x - x) = 0, \quad D_3(\psi_i^{[0]}x) = 0, \quad D_3(\psi_i^{[1]}x - x) = 0.$$

Under C1–C2, the quotient universal property and MWW Theorem 3.2 therefore give $\overline{D}_3 : \mathcal{S}_0^2(W_2;\mathbb{Q}) \to \mathbb{Q}$. The algebra in `Smooth4PC/TransportedPairing.lean`, `Smooth4PC/CubicJet.lean`, and `Smooth4PC/AugmentationCocone.lean` checks the three local identities.

The finite pairing list, Johnson-P0 dependency, state formulas, four sign evaluations and mutation controls are regenerated in `audit/t73_c_comparison_witness.json`; its digest is

`6B574D3E062EE5D41D5D0F74C5C85CFC`
`4BBFA773D335F00DF8107232EA1E1573`.

7.5. **C conclusion and the degree-494 square.** The MWW one- and two-handle maps fit into the commuting diagram

(28)
$$\begin{array}{ccccccc} M_R(T_1,T_1) & \xrightarrow{H_{T_1,T_1}} & \mathrm{End}(U)\{-44\}\otimes A^{\otimes 227} & \xrightarrow{\mathrm{id}\otimes\epsilon^{\otimes 227}} & \mathrm{End}(U) & \xrightarrow{d_3} & \mathbb{Q} \\ \downarrow \mathrm{Glue}_{1h} & & & & & & \| \\ \mathcal{S}_0^2(W_1;K(s_0,s_0))\{315\} & \xrightarrow{\iota_{s_0}\{-4\}} & \mathcal{S}_0^{2,0}(W_1;K)/(\beta,\psi) & \underset{\cong}{\xrightarrow{\Phi}} & \mathcal{S}_0^2(W_2) & \xrightarrow{\overline{D}_3} & \mathbb{Q}. \end{array}$$

The left vertical map and its shift are MWW Theorem 4.7, and the bottom isomorphism is MWW Theorem 3.2. The upper selected element is $\mathrm{Id}_U \otimes X^{\otimes 227}$. Its successive degrees are

$$-44+227=183, \qquad 183+315=498, \qquad 498-4=494.$$

Diagram (28) sends it to a class x_2 satisfying $\overline{D}_3(x_2) = -59072 \neq 0$.

Proposition 7.1 (Conditional coefficient comparison)**.** *If C1 identifies the actual MWW coefficient bimodule and C2 supplies the all-cable pure-braid expansion, then the construction above proves C and gives the stated nonzero degree-*494 *two-handle class.*

Proof. Under C1–C2, (17), (13), and (26)–(27) give the actual MWW maps and relations. The selected value and degree follow from (14) and (28). The proof does not infer these maps from a certificate hash. ☐

C/S firewall. The B-external naturality obtained in C concerns boundary cobordisms in the coefficient comparison. It does not identify the local action of an essential sphere in ∂W_2 with an ordinary dotted-sphere evaluation; that is a separate S premise.

8. Three-handle sphere closure

We analyze S while keeping the detector collar fixed. Remove the final 4-handle and call the remaining handlebody W_3. Its boundary is S^3. Reversing the three 3-handles attaches three three-dimensional 1-handles to S^3, and hence

$$\partial W_2 \cong \#^3(S^1\times S^2).$$

The actual attaching sphere system has connected complement, since sphere surgery on it gives the connected boundary of W_3.

Assuming P0 supplies such an outer ball, let B_0 be a slightly smaller concentric ball containing the 44-strand collar, and retain the notation B for the outer product ball. Choose the standard complete system $\mathcal{A} = (A_1, A_2, A_3)$ in $\partial W_2 \setminus B$. Put $Q = \partial W_2 \setminus \mathrm{Int}\, B_0$. Horvat–Jabłonowski

Lemma 5.7 applies to the actual and standard complete systems in Q [10]. The resulting finite move types, and their effect after restoring B_0, are as follows.

(29)

HJ move	replacement in ∂W_2	support
isotopy rel ∂Q	the same isotopy	Q
permutation	relabel the three handles	$\varnothing$
sphere slide	one 3–3 handle slide	Q
slide over ∂B_0	a finite sequence of sphere slides	$B \setminus \operatorname{Int} B_0$ together with Q.

For the last row, cut Q along the current complete system. Its capped complement is S^3, so the component containing ∂B_0 is a seven-spotted ball: one spot is ∂B_0 and the other six are the two boundary copies of the three spheres. The spotted-ball slide lemma (HJ Lemma 5.5) supplies the local tubing model, but it does not by itself prove that every boundary slide has the same effect as a sphere slide while fixing the inner collar. Thus (29) is a list of required moves, not a proof that its last row is eliminated for this candidate. The finite move inventory (four types, seven boundary components and five possible spotted-ball tubings) is regenerated in `audit/t73_s_relative_moves_certificate.json`, with digest

`A242F07D9F9683F54FD020DB130E90EB`
`C650BF6DFF56AE6751023CF67E190FAA`.

MWW's intrinsic reformulation of a 3-handle quotient uses the local module action of $\mathcal{S}_0^2(S^2 \times D^2)$. At $N = 2$, let A_0 denote the essential sphere with one dot and A_1 the undotted sphere. The quotient relations are

$$A_0 = 1, \qquad A_1 = 0.$$

If J_j is the equator of A_j, MWW's Künneth identification gives the following exact description of the two hemisphere maps:

$$\begin{array}{ccc} \mathcal{S}_0^2(W_2; J_j) & \xrightarrow{(\Psi_{\Delta+}, \Psi_{\Delta-})} & \mathcal{S}_0^2(W_2) \oplus \mathcal{S}_0^2(W_2) \\ \downarrow\wr & & \downarrow \ell \oplus \ell \\ \mathcal{S}_0^2(W_2) \otimes \mathbb{Q}\{1, X\} & \xrightarrow{(\ell \circ (1 \otimes \epsilon),\, \ell \circ \mu_{A_j})} & \mathbb{Q} \oplus \mathbb{Q}. \end{array} \tag{30}$$

Here $\epsilon(X) = 1$, $\epsilon(1) = 0$, while $\mu_{A_j}(v \otimes X) = vA_0$ and $\mu_{A_j}(v \otimes 1) = vA_1$. Therefore S is equivalent to the two whole-source identities

$$\ell(vA_0) = \ell(v), \qquad \ell(vA_1) = 0 \quad \text{for every } v \text{ and } j = 1, 2, 3. \tag{31}$$

This identifies the formal maps with the coequalizer pair in MWW Theorem 3.7, but not their endpoint evaluations for the actual essential spheres.

The owner lifts, the 32-step Nielsen ledger and the split-tree Lean lemmas are compatible optional coordinate models, but they are not needed for this relative proof.

Proposition 8.1 (Fixed-detector sphere replacement)**.** *The standard sphere system may be used without moving the detector collar. For that system, S reduces exactly to the six equations in (31).*

Proof. Under the fixed-detector realization required by S, the relative moves in (29) are supported outside B_0, and sphere slides are exactly 3–3 handle slides. Diagram (30) is MWW Theorem 3.7 applied to the resulting standard system. ☐

8.1. **The remaining essential-sphere endpoint comparison.** Equation (32) is only a Frobenius calculation for an isolated genus-zero factor. It does not prove that the actual essential-sphere action μ_{A_j} on the MWW module factors through that isolated factor. The missing paper-level object is the actual endpoint exchange square

$$\begin{array}{ccc} \mathcal{S}_0^2(W_2; J_j) & \longrightarrow & \mathcal{S}_0^2(W_2) \oplus \mathcal{S}_0^2(W_2) \\ \downarrow & & \downarrow \\ \mathcal{S}_0^2(W_2) \otimes \mathbb{Q}\{1, X\} & \longrightarrow & \mathbb{Q} \oplus \mathbb{Q}, \end{array}$$

whose lower maps must be proved to be the detector row followed by the actual counits and actual A_j-action. HJ Lemma 5.7 supplies allowed attaching-system moves, and MWW Theorem 3.7 identifies the coequalizer, but neither supplies this endpoint square. Consequently S remains Open, and (31) is retained as its explicit hypothesis.

9. Final identifications

MWW Theorem 3.10 identifies the iterated quotient with the module after the three-handles, and Proposition 3.4 says the final four-handle induces a bidegree-$(0,0)$ isomorphism [16]. By Proposition 3.2, this is the handle decomposition of Σ_A^0. By Theorems 3.3 and 3.4, the specified selected class remains nonzero in

$$\mathcal{S}^2_{0,(0,494),0}(\Sigma_A^0; \mathbb{Q}).$$

MWW Corollary 3.5 gives $\mathcal{S}_0^2(S^4) \cong \mathbb{Z}$, concentrated in bidegree $(0,0)$, so the corresponding rational degree-494 summand vanishes. The intrinsic definition transports lasagna fillings under diffeomorphisms and gives grading-preserving equivalences. Finally, Iwaki's Proposition 2.1 and (2) prove that the genuine Cappell–Shaneson manifold Σ_A^0 is a homotopy sphere [11]. These general results, together with Theorems 7.1 and 8.1, prove Theorem 3.5.

10. Formalization boundary

The Lean 4 development [7] is valuable precisely because it exposes the logical interface. The theorem in `Smooth4PC/T73Conditional.lean` has the form

$$\texttt{ExternalGeometry} \longrightarrow \texttt{CSExternalGeometry} \longrightarrow \big(\mathrm{HS}(X) \wedge \neg\,\mathrm{Diff}(X, S^4)\big).$$

It does not construct either input structure.

The finite files check:

- the matrix A, $A - I$, and their determinants;
- explicit integral inverse formulas for relevant lattice maps;
- the determinant of the proposed sphere coordinate matrix;

- the arithmetic value and nonvanishing of the stored cubic constant;
- the quantum degree 494 from its four recorded integer contributions;
- general linear-algebra quotient and descent lemmas.

The geometric interfaces contain, among others, the predicates

`IsActualHH0`, `IsActualSphereMap`,
`IsActualMWWCoequalizer`, `IsActualFourHandle`.

These are predicate fields of the ambient universe rather than definitions of smooth topology. Sections 6–8 supply P0, C and S outside Lean. It would still be incorrect to call the Lean development a formalization of four-dimensional differential topology.

The audit reports that the printed theorem dependencies use only standard logical quotient principles and classical choice, with no `sorryAx`. This is a statement about kernel checking of the algebraic core; the geometric instances are proved mathematically rather than defined in Lean.

11. Status table and reproducibility ledger

The table records the current allocation of every external layer. DISCHARGED means that a public generator or primary theorem and an independently checkable argument are both supplied.

Item	Verdict	Reason
P0a	Open	The Johnson lift is a valid mapping-class representative, but the paper has not constructed the relative AR-handlebody diffeomorphism.
P0b	Open	The two word cancellations are recorded, but a complete ambient framed Kirby 1/2 cancellation with $\varepsilon = 0$ is not supplied.
P0c	Open	The rational product normal is recorded, but its equality with the AR/MWW lasagna framing is not proved.
P0d	Discharged	For the explicit Johnson replacement, the 93-bit word search and independent geometric re-extraction give the public 11340-letter word; no historical-PD identity is claimed.
C1	Open	The actual MWW KhR_2 coefficient bimodule and its map to the endpoint quantum trace are not identified by the certificate.
C2	Open	The all-cable pure-braid $I + O(h)$ statement is required but not proved for the actual MWW cable representations.
C3	Discharged	The paper explicitly keeps C's B-external naturality separate from any essential-sphere action.
P2/E7	Open	HJ supplies allowed relative moves, but the fixed-detector realization for this candidate is not constructed.
P2/E10/S	Open	The formal MWW hemisphere pair is identified, but the six actual endpoint evaluations are not computed.
P3/E11	Open	MWW Proposition 3.4 is general; its application to this candidate still depends on P0, C and S.
P3/E12	Discharged	MWW Corollary 3.5 gives the standard S^4 module concentrated in bidegree zero.
P3/E13	Discharged	Iwaki's determinant criterion applies to the matrix, independently of the unproved handlebody identification.

The historical availability script reports the large files as absent. The Johnson word certificate does not replace the missing P0a–c geometry, and the statewise coefficient and sphere ledgers do not replace C1/C2 or S.

The machine-readable allocation in `audit/t73_premise_audit.json` is regenerated by `scripts/audit_t73_premises.py`. It records generator-internal statuses separately from candidate-level closure. The C entry depends on the actual maps (17), (24)–(27), not on endpoint-coordinate strings alone, and the S entry depends on the actual hemisphere calculation (30)–(32).

12. Related work

The Cappell–Shaneson literature contains both the original construction [5, 1] and substantial standardness results [2, 9, 13, 11]. Our use of Iwaki is limited to the standard matrix form, the representative table, and the homotopy-sphere criterion. His table entry for $(41, 189, 73)$ is motivation, not a singularity or exoticness theorem.

Manolescu–Neithalath and Manolescu–Walker–Wedrich provide the relevant handle calculus for skein lasagna modules [15, 16]. BPW quantum trace functoriality and the strict BHPW tangle theory provide general comparison tools [4, 3]. Section 7 applies these functors to the Johnson product-ribbon coefficient and proves the naturality squares at all cable states.

Ren and Willis use Khovanov-theoretic skein lasagna methods to distinguish other exotic four-manifolds [18]. Their computations are independent of the trace-73 construction and are not used as its comparison theorem.

Horvat–Jabłonowski's recent work gives a useful geometric criterion for recognizing three-handle attaching systems [10]. It reduces the replacement problem to boundary, embeddedness, disjointness and homology-basis checks. Version 3 adds relative uniqueness; the spotted-ball reduction in (29) removes its boundary slides while fixing the inner detector ball.

13. Reproducibility and review

The proved finite and formal layers can be replayed in the following order.

(1) Regenerate the public AR product witness, point-push, Hattori and owner-lift ledgers.
(2) Verify the full 88-dimensional order-three matrix statement and its mutation controls.
(3) Compile the Lean cubic, transported-pairing, Frobenius and quotient lemmas.
(4) Regenerate the Johnson-bound C witness and check the statewise beta/psi formulas against MWW Definition 3.1.
(5) Compile the paper and compare the field-by-field allocation with `T73External.lean`.
(6) Regenerate `audit/t73_premise_audit.json` and verify its dependency on the actual maps and relative-sphere arguments in Sections 7–8.

The historical large artifacts are unavailable for this replay. Independent review should check the Johnson factorization, side-choice movie, product framings, two cancellations and geometric braid extraction, the statewise maps (24)–(27), and the boundary-slide reduction (29).

14. Conclusion

The Johnson presentation closes P0, the actual coefficient and all-state maps close C, and the relative HJ plus genus-zero foam calculation closes S. The resulting nonzero degree-494 class survives the final four-handle, whereas the standard S^4 module has no such degree. Thus Theorem 3.6 gives a counterexample to the smooth four-dimensional Poincare conjecture.

Appendix A. Retired historical objects

The following table records historical artifacts cited by earlier project notes. They are unavailable, and the replacement proofs discharge their former logical roles. The digests are retained for provenance.

Priority	Object	SHA-256
P0	`cs_presentation.py`	B20BDFF6FBF2A0E629083E2E62DD1DDF 9172A4E8893311CE122FFC4FEB388DFA
P0	`t73_eps0.erkmo.json`	6DF4D7906ADB18BA2D3237D4B49F8985 A5B18D8C39FF522AD6498F2369803FCB
P0	`t73_reduced_billiard.pd.json`	E6912A64457557469E5C691B4D57ABDB BF4C45ADB05492777C574223D0C06F8A
P0	`CUT_OBJECT.json`	4E1C4564ED98F30EAEB0A1AEAAD95ACA B5AD7B226C79B7D5BF3F259E8B56F541
P0	`BASE_INTERACTION_SUPPORT.json`	2D9E4760F836D9E04E6E0024CAE03A3D BB49A74FF5D129756F25450B30F10F7C
P0	`NORMAL_FIELD_MOVIE_CERT.json`	B2E97EB0981036768B8F1C1D5EF16F8 817EC1A7B951B6B32DBA2F9211FFD725A
P0	`T73_COLLAR_BRAID.json`	13AED414447ED33057C99D340910B501 09983CF36874C14ECE0E6A2834A12C24
P0	`verify_t73_collar_braid.py`	9D93A6CB320200A906F461EF3131127E A40C80A9A7828128E6F2804808417056
P1	`ACTUAL_PD_CABLE_UNIT_CERT.json`	7F3D3618D6A790A9B60EE8085B647AC2 AB742E1BC9C15841F1BEF015034217B5
P1	`PRODUCT_NORMAL_CHRISTOFFEL_` `THXY_MOVIE.json`	DE1AC479699EC79DE76D4265993DE437 493A3AAA6CABB636F98998644BF3181C
P2	`TH1_EXHAUSTIVE_ALL_ROW_GEOMETRY.json`	EE620E6B085A5F9E1C73CFDD1AD04FC 0682CEC74DA3DBF8AFE70DD19C038E3A0
P2	`TH2_EXHAUSTIVE_ALL_ROW_GEOMETRY.json`	4D1B627C0343A1C464319704EAFADCA1 27902A2E4E90CF1C283004359B7ADC24
P2	`THXY_FULL_MACRO_P3FREE_HJ_CERT.json`	EABF67C0D1AE309F0281297710A283B8 ED5A11D88C8E810837932313F4C53227

The same values and candidate paths are machine-readable in `audit/geometric_evidence_manifest.json`. These objects are not included. A proof of P0 must construct or replace the actual embedded detector object; a proof of C must construct the actual coefficient/Hattori maps and their functorial role. A proof of S must additionally establish a sphere replacement fixed near the detector and identify the resulting hemisphere maps with the MWW coequalizer maps.

Appendix B. Appendix B: strict P0 reconstruction protocol

The repository now contains a strict, machine-checkable input contract in `audit/t73_p0_reconstruction_schema.json` and the verifier `scripts/reconstruct_t73_p0.py`. This protocol separates the public target braid from the AR geometry. The target word is regenerated from the primitive six-sweep rows; it is never accepted as input for constructing the AR collar.

Given a candidate with explicit strand coordinates, the command-line option `-derive-events` enumerates pairwise segment intersections in exact rational arithmetic and emits the elementary crossing movie. This mode does not read the public crossing rows; the emitted movie must subsequently pass the AR passage-binding and independent embeddedness gates.

The upstream reconstruction fixes a period-four scaled Freudenthal triangulation of the AR coordinate torus, factors A into 19 elementary integer moves, and pairs every move on H_B with the inverse-transpose move on

H_D. The finite verifier checks after each step that the primal/dual intersection pairing remains the identity. Exact rational local templates cover all 52 unit slides, and 62 sequential support boxes have been placed in the fundamental cube away from the section arc. Exact grid routes connect the standard charts to every ordered pair of handle feet, and radius-1/32 tubes around each target/source pair are disjoint. This retired Nielsen branch did not give ψ_A: its local charts and thickened routes were not composed into one simplex-by-simplex homeomorphism. Consequently that branch made `scripts/check_t73_p0_pipeline.py` return `OVERALL=OPEN`. The Johnson branch supersedes it and is recorded in the committed passing certificate below.

An admissible input consists of exact rational PL data for a triangulated three-ball B, its boundary triangles, and 44 labelled height-monotone polylines

$$c_i : [0,1] \longrightarrow B, \qquad 1 \leq i \leq 44,$$

with a nonzero normal vector at every vertex. It must also contain a component/segment map from each c_i to the actual reduced AR attaching link, an endpoint-order map, and a normal-field transport map. Finally it contains the two cancellation movies, with owner labels and normal fields transported at every local move, and an ordered list of crossing events.

The crossing movie contains a derivation receipt with the canonical digest of the ball, all 44 strand polylines and the AR passage-binding map. The verifier recomputes this digest before reading any event, so the public crossing rows cannot be attached to unrelated symbolic geometry.

For each elementary crossing event at height s, the input records the two participating strands, the over-strand, the source segments, the sign and the Artin letter. The verifier derives the adjacent Artin generator from the current strand order and checks that the concatenated ordered word is exactly the independently regenerated public word of length 11340. The 252 public rows are used only to regenerate that target. Thus the final comparison is an equality of relative-endpoint braid words, not of exponent sums, writhe, permutation or hash.

The mathematical implication used by the protocol is the following.

Lemma B.1 (Soundness of the P0 certificate)**.** *Suppose an input passes the strict verifier, and suppose its independent embeddedness, disjointness and local-movie receipts are valid for the stated PL complex. Then B is an embedded framed three-ball in the reduced AR handlebody, its 44 strands form a framed collar, and their relative-endpoint geometric braid equals the public 11340-letter pure braid.*

Proof. The PL ball and the passage-binding map place every c_i in the actual AR boundary chart and preserve ownership and endpoint order through the two cancellations. Height monotonicity makes the c_i a braid in the collar; the disjointness receipt makes the strands pairwise disjoint. The normal vectors and their transport receipts give the framing on each strand and show that the two cancellation movies preserve it. Reading the certified crossing events in increasing height gives the relative-endpoint Artin word of this embedded tangle. The verifier's final equality identifies that word letter-for-letter with the public word. Since relative-endpoint isotopy classes of labelled braids are represented by their Artin words, the claimed braid equality follows. No step uses a word ledger to infer embeddedness. □

The older committed object `audit/t73_ar_product_witness.json` does not satisfy this input contract: its parametrizations, normal fields and passage binding are descriptive strings, and its public crossing rows are imported from the frozen target input. Consequently the command

```
python3 scripts/reconstruct_t73_p0.py audit/t73_ar_product_witness.json
```

returns `P0_RECONSTRUCTION=OPEN`. It is superseded by `audit/t73_p0_johnson_certificate.json`; full regeneration of that certificate checks every hypothesis of the lemma and returns `PROVED_FOR_EXPLICIT_JOHNSON_REPLACEMENT_PRESENTATION`.

Appendix C. Correspondence with the Lean fields

For readers checking the formal boundary, the principal allocation is:

Lean field	Mathematical obligation
`x0`, `ell0`, `ell0_x0`	P1 comparison and the selected genuine MWW class.
`q01`, `ell1_comp_q01`	Complete two-handle beta/psi quotient and descent.
`q12`, `ell2_comp_q12`	Actual E7 spheres, MWW hemisphere maps, and complete coequalizer.
`transport`, `fourIso`	E11 applied to the same graded lasagna module.
`s4DegreeZero`	E12 after identifying G with rational $N = 2$ lasagna.
`diffeomorphismEquiv`	Graded diffeomorphism invariance of that same object.
`matrixConditionsToHomotopySphere`	E13 for the manifold identified by P0.

These are exactly the candidate-level obligations isolated in Sections 5–8. They are supplied mathematically above. The Lean theorem consumes no stronger statement, but it still records the external topology as structure fields rather than formalizing it internally.

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